

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\mathcal{L} = \frac{1}{4}(F_{\mu\nu}^a)^2 + \overline{\psi}_\mu^i \left[\not{D}^{ij} \psi_\mu^j - \frac{1}{2} \gamma_\alpha D_\mu^{ij} \psi_\alpha^j - \frac{1}{2} \gamma_\mu D_\alpha^{ij} \psi_\alpha^j + \frac{3}{8} \gamma_\mu \not{D}^{ij} \gamma_\alpha \psi_\alpha^j \right].$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\mathcal{I}(\delta) = \inf_{\lambda \in \mathbb{R}_+^2} \{ \langle \lambda, \delta \rangle + \mathcal{I}^*(\lambda) \} = \inf_{\lambda \in \mathbb{R}_+^2} \left\{ \langle \lambda, \delta \rangle + \sup_{\gamma \in \Pi(\mu_1, \mu_2)} \int_{\mathcal{V}} f_\lambda(v) d\gamma(v) \right\},$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\left(\phi_1(x), \begin{pmatrix} \phi_2(x) \\ \phi_3(x) \end{pmatrix}, \begin{pmatrix} \phi_4(x) \\ \phi_5(x) \end{pmatrix} \right) \rightarrow \left(c e^{\lambda x} \phi_1(x), U \begin{pmatrix} e^{-\lambda' x} \phi_2(x) \\ e^{-\lambda' x} \phi_3(x) \end{pmatrix}, V \begin{pmatrix} e^{\lambda'' x} \phi_4(x) \\ e^{\lambda'' x} \phi_5(x) \end{pmatrix} \right)$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\frac{\partial L}{\partial w_{ijkl}}(g_{ij}, g_{ij,k}, g_{ij,kl}) = \frac{\partial x^i}{\partial \tilde{x}^a} \frac{\partial x^j}{\partial \tilde{x}^b} \frac{\partial x^k}{\partial \tilde{x}^c} \frac{\partial x^l}{\partial \tilde{x}^d} \frac{\partial L}{\partial w_{abcd}}(\tilde{g}_{ij}, \tilde{g}_{ij,k}, \tilde{g}_{ij,kl}).$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$q^Q \prod_{k=1}^{Q-1} \frac{qz_i - z_k}{z_i - qz_k} = \frac{\left(z_i - i\tau\kappa\lambda^{-\frac{1}{2}}q^{\frac{1}{2}} \right) \left(z_i + i\kappa\lambda^{\frac{1}{2}}q^{\frac{1}{2}} \right)}{\left(q^{\frac{1}{2}}z_i + i\tau\kappa\lambda^{-\frac{1}{2}} \right) \left(q^{\frac{1}{2}}z_i - i\kappa\lambda^{\frac{1}{2}} \right)}.$$